Commercial Excellence with AI: market whitespace, sales performance and an automated business review

PUBLIC + SIMULATED DATA

A case study for an international cardiovascular device business in EMEA and APAC. The real market size (19 countries, 1,612,229 coronary angioplasties a year, OECD) is combined with simulated sales of six generic product lines, analysed with SQL, Python and Power BI, then reported automatically by Claude with a check on every number.

REAL + SIMULATED · PAGE 1

EUR 33.6 m

untapped potential a year. Türkiye and Germany hold 65% of it: big markets where our share is 3.3% and 5.8%, vs an 8.5% benchmark.

SIMULATED · PAGE 2

-5.4%

year to date vs target (EUR -5.5 m). France and Germany explain 73% of the gap; stent price erosion alone cost EUR 2.0 m.

SIMULATED · PAGES 3-4

10 hospitals

flagged at risk (-61% in 3 months). The 2026 forecast ends 5.3% below target, and even the best scenario misses.

AI · PAGE 5

29/30

test questions answered correctly by text-to-SQL. 0 of the 1,670 numbers in 30 reviews written by Claude could not be traced back to SQL.

Business questions

1 · Market Where is the potential in EMEA and APAC?
2 · Performance What explains the gap to target?
3 · Customers Which hospitals carry growth, and which may leave?
4 · Forecast Will we hit the annual target?
5 · AI How can AI save the team time without inventing numbers?

Recommendation: defend first: win back the 10 at-risk hospitals and hold stent prices in tenders; then invest where big markets have a low share (Türkiye, Germany); automate the monthly review to free about 46 analyst hours a month for that work.

Data & methods

REAL OECD Health Statistics: coronary angioplasties by country (latest year 2024, Türkiye 2025, New Zealand 2023); World Bank population, GDP and health spending.
SIMULATED sales of 357 fictional hospitals, 6 generic product lines, Jan 2023 - Oct 2026, with targets; market shares of 2-25% on the real market size.
SQL whitespace, sales vs target, price-volume-mix bridge, account alerts with window functions (DuckDB).
Python seasonal forecast checked on 6 months it never saw.
AI Claude API: monthly review written from SQL facts with a number check; text-to-SQL on a read-only database; tested on 40 questions.
Power BI star schema (3 facts, 4 dimensions), 68 DAX measures.

How to read this report

Titles state the conclusion of the page. **Year to date** January-October 2026 vs the same months of 2025. **Blue** = data, **grey** = comparison, **red** = below target or at risk; dashed lines mark targets and thresholds. **Labels** REAL, ESTIMATE and SIMULATED are kept apart on every page.

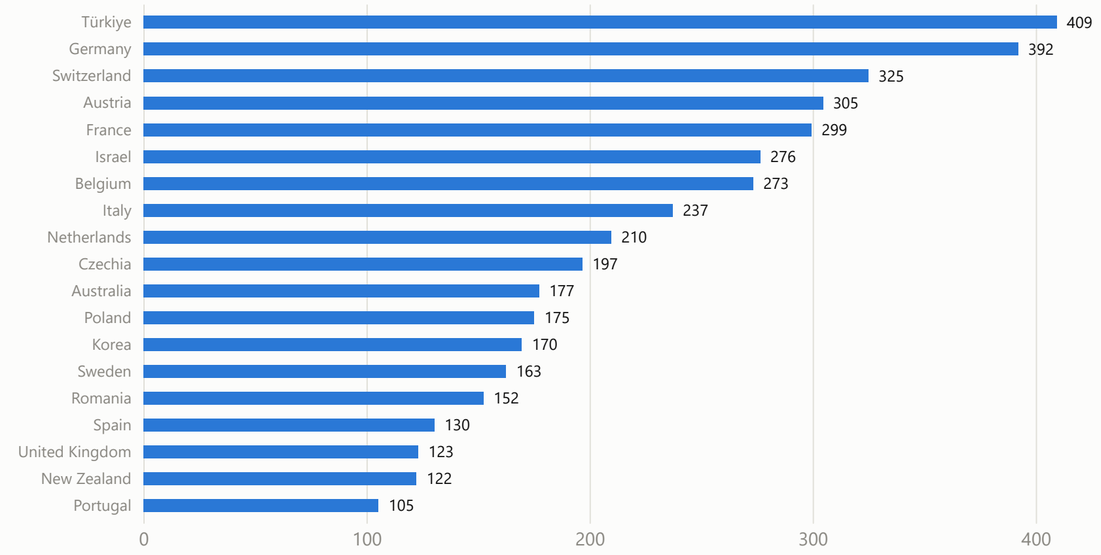
How it was built I used Claude Code to write the scripts efficiently. I made the decisions: the business questions, which data to use and how to simulate it, the methods, the checks and the page design, and I reviewed every result.

Türkiye and Germany hold 65% of the EUR 33.6 m untapped potential

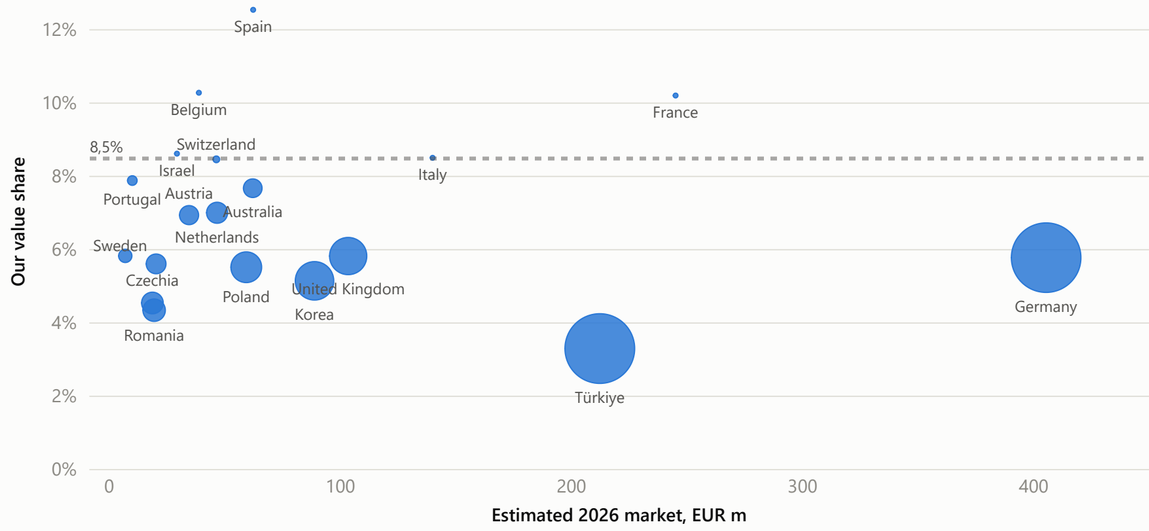
PUBLIC + SIMULATED DATA

Real angioplasty rates range from 105 per 100,000 people (Portugal) to 409 (Türkiye, +7.6% a year since 2019). Both are big markets where our share is low. Potential = estimated market x benchmark share (8.5%, our top quartile of countries) - our last-12-month revenue. Market value is an estimate (real procedures x assumed devices and prices).

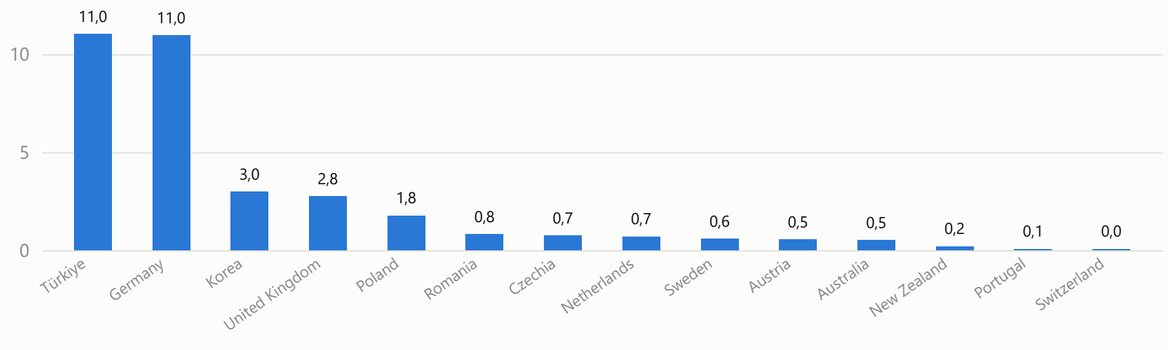
REAL · Coronary angioplasties per 100,000 inhabitants, latest OECD year



Market size vs our share (bubble = untapped potential; dashed = 8.5% benchmark)



Untapped potential by country, EUR m a year (countries at or above the benchmark have none)



Where to invest: market, share, potential and real procedure growth since 2019

Country	Market 2026 (est.)	Share	Potential	Decision	PCI growth p.a.
Türkiye	212 519 594	3,3%	11 017 223	Invest: big market, low share	7,6%
Germany	405 614 500	5,8%	10 966 336	Invest: big market, low share	-1,3%
Korea	89 090 363	5,1%	2 972 265	Invest: big market, low share	2,5%
United Kingdom	103 653 786	5,8%	2 757 203	Invest: big market, low share	0,5%
Poland	59 563 668	5,5%	1 766 187	Invest: big market, low share	6,0%
Romania	19 702 155	4,3%	815 115	Develop selectively: small market, low share	1,5%
Czechia	18 961 723	4,5%	747 150	Develop selectively: small market, low share	-1,7%
Netherlands	46 942 184	7,0%	694 072	Invest: big market, low share	-1,0%
Sweden	20 559 017	5,6%	590 801	Develop selectively: small market, low share	-3,9%
Austria	34 811 064	6,9%	537 182	Develop selectively: small market, low share	-0,7%

Real: OECD Health Statistics, surgical procedures CM36_TRS (coronary angioplasty); World Bank WDI · Market value: estimate · Simulated sales, targets and hospitals (src/generate_sales.py, seed 42), calibrated on real OECD market size · Independent case study · Not affiliated with Cordis

Revenue is 5.4% below target year to date: France and Germany explain 73% of the EUR 5.5 m gap

SIMULATED DATA

Revenue still grew +1.3% on last year: volume added EUR 3.8 m and mix EUR 1.4 m (more drug-coated balloons, +21% units), but price took EUR 4.0 m. Coronary stents lost 7.2% of price (51% of the price effect), Germany EUR 1.2 m after its 2026 stent tender. The bridge closes exactly.

Revenue Jan-Oct 2026, EUR m

95,4

Target Jan-Oct 2026, EUR m

100,8

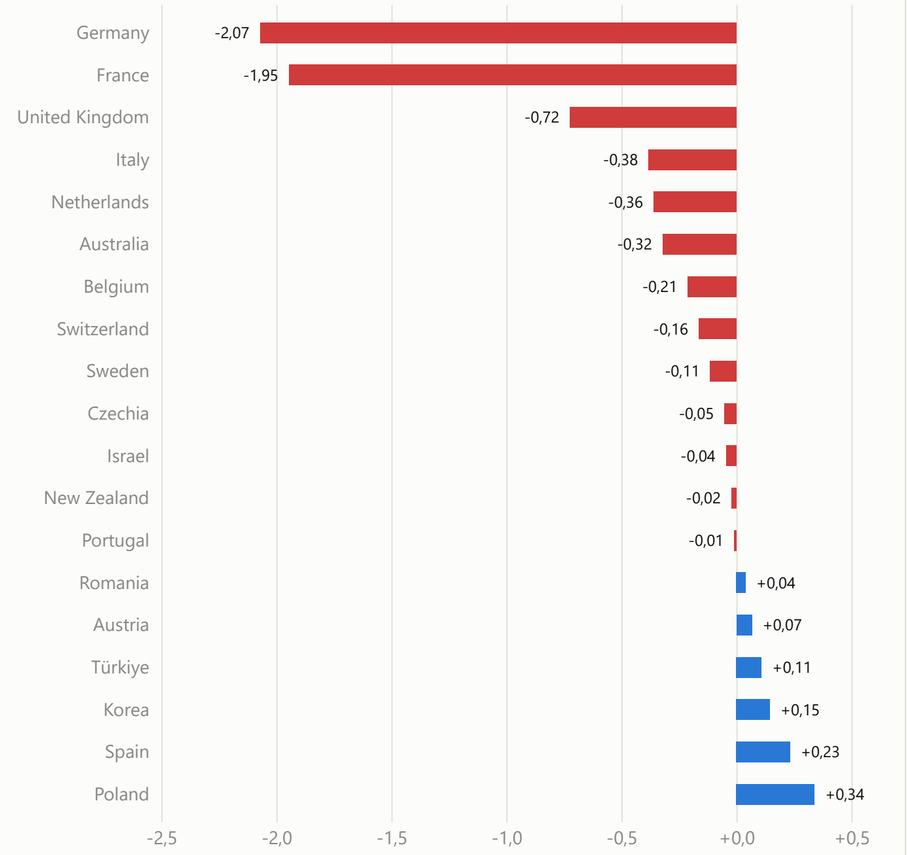
Gap to target, EUR m

-5,48

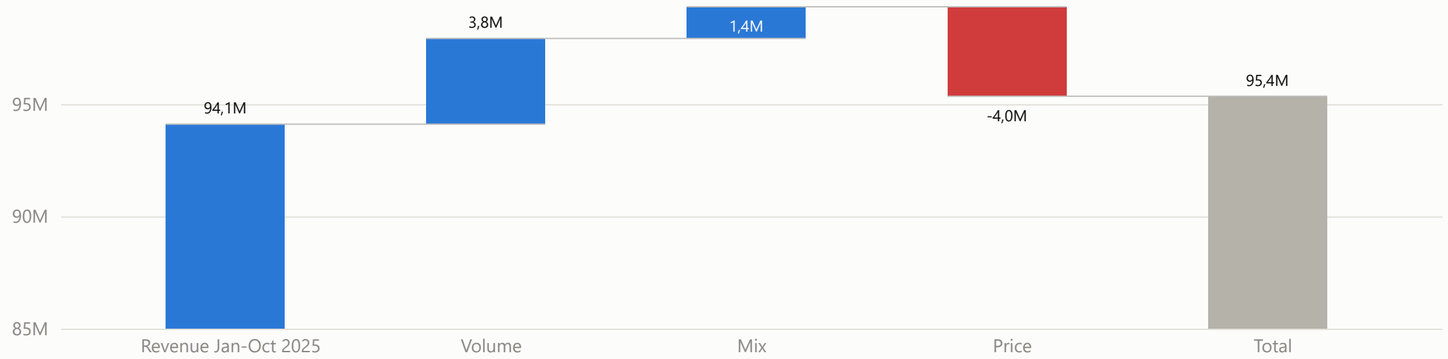
Growth vs Jan-Oct 2025

+1,3%

Gap to target by country, EUR m (red = below target)



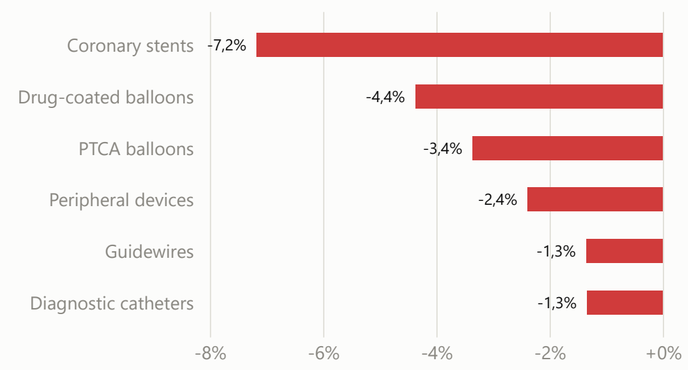
Price-volume-mix bridge, Jan-Oct 2025 to Jan-Oct 2026 (last bar = total 2026), EUR



Bridge by country (volume + mix + price = change)

Country	2025 YTD	Volume	Mix	Price	Change
Germany	20 363 873	100 250	179 453	-1 193 319	-913 616
France	21 507 534	-636 896	560 595	-646 974	-723 275
United Kingdom	5 414 885	-150 487	-112 794	-152 736	-416 017
Netherlands	2 940 065	-125 765	-19 394	-78 846	-224 006
Sweden	1 027 908	-66 560	16 850	-26 604	-76 314
Australia	4 062 820	38 356	39 111	-126 996	-49 529
Belgium	3 380 899	-51 548	155 378	-121 381	-17 551
New Zealand	355 218	-2 488	8 291	-10 975	-5 172
Czechia	722 452	13 593	5 014	-22 961	-4 354
Portugal	650 292	14 329	26 817	-20 842	20 305

Net price change by product line, vs Jan-Oct 2025



Simulated sales, targets and hospitals (src/generate_sales.py, seed 42), calibrated on real OECD market size · Independent case study · Not affiliated with Cordis

10 hospitals are at risk: their last 3 months are 61% below a year earlier

SIMULATED DATA

The top 20% of hospitals make 60% of revenue, so a few lost tenders move the year: the flagged accounts (3 in France) have lost a run-rate of about EUR 5.6 m a year. Alert = last 3 months at least 30% below the same months last year, on accounts worth EUR 30 k+: it caught 10 of the 10 losses built into the simulation, 0 false alarms. New hospitals since 2025 (35, 12 in Türkiye) bring only 2.8% of revenue so far.

Hospitals buying (last 12 months)

347

Revenue share of the top 20%

60%

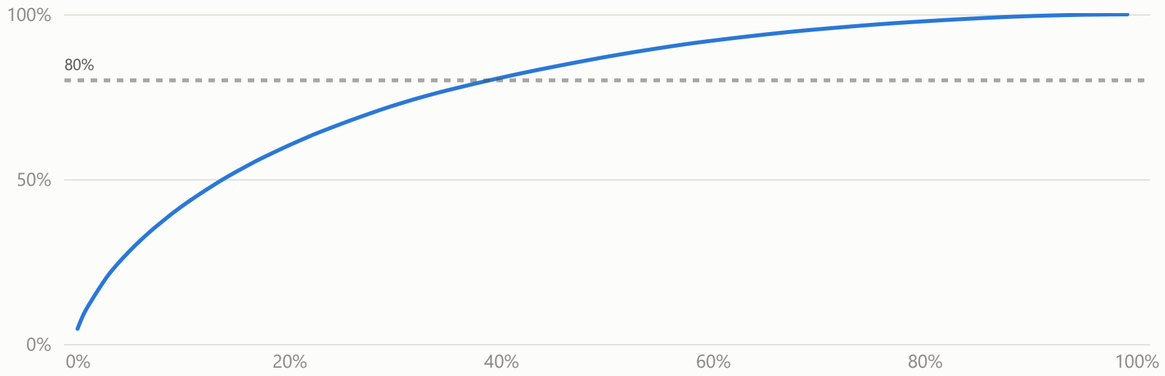
Hospitals flagged at risk

10

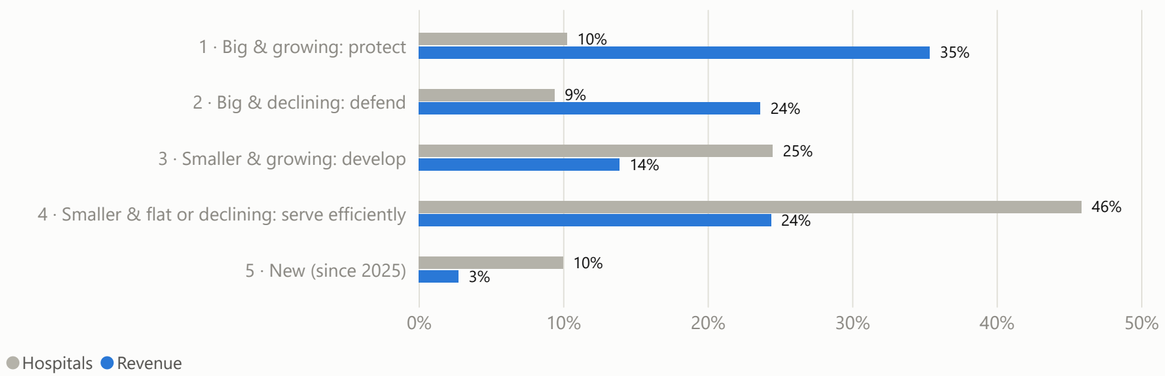
Revenue share of new hospitals

2,8%

Cumulative share of revenue by hospital rank (Pareto curve, dashed = 80%)



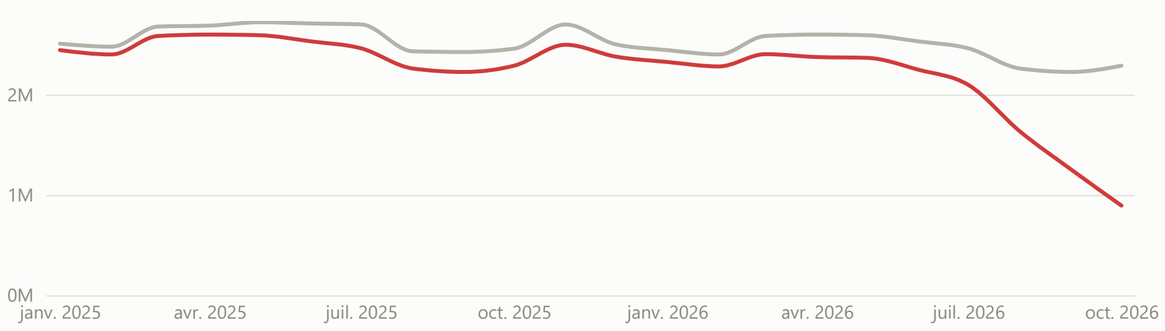
Segments: share of hospitals (grey) vs share of revenue (blue), last 12 months



Watch list: hospitals at risk, largest first (call before the next order cycle)

Hospital	Country	Type	Aug-Oct 2025	Aug-Oct 2026	Change	Segment
FR-054	France	University hospital	461 794	123 030	-73%	2 · Big & declining: defend
DE-012	Germany	Public general hospital	292 494	76 435	-74%	2 · Big & declining: defend
GB-002	United Kingdom	Public general hospital	284 521	28 800	-90%	2 · Big & declining: defend
IT-035	Italy	University hospital	263 206	181 335	-31%	2 · Big & declining: defend
AU-007	Australia	Private hospital	240 258	146 589	-39%	2 · Big & declining: defend
NL-001	Netherlands	Public general hospital	212 010	116 823	-45%	2 · Big & declining: defend
FR-033	France	Public general hospital	186 985	62 021	-67%	2 · Big & declining: defend
FR-029	France	Public general hospital	180 385	82 695	-54%	2 · Big & declining: defend

The flagged hospitals together: rolling 3-month revenue (red) vs a year earlier (grey)



Simulated sales, targets and hospitals (src/generate_sales.py, seed 42), calibrated on real OECD market size · Independent case study · Not affiliated with Cordis

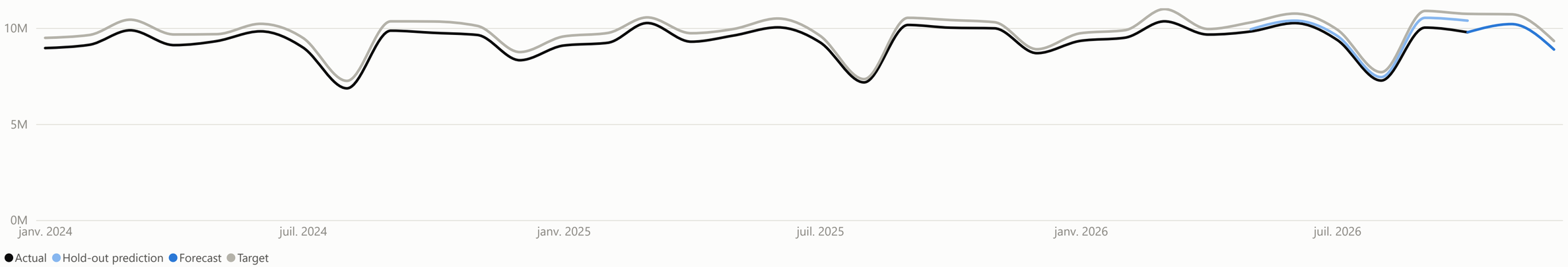
2026 will end about EUR 6.4 m (5.3%) below target: even the best scenario misses

SIMULATED DATA

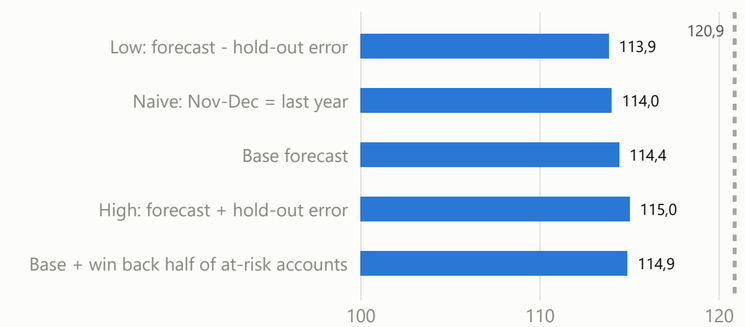
Forecast per country = same month last year x trailing 12-month growth. Checked on May-Oct 2026, hidden from the model: 3.1% monthly error. It over-forecast by EUR 1.75 m, almost exactly what the at-risk hospitals lost (EUR 1.76 m): the model cannot see lost tenders, the account alert can. Reaching target would need +34% on the November-December forecast.

2026 forecast, EUR m	2026 target, EUR m	Forecast gap to target, EUR m	Monthly error on 6 hidden months
114,4	120,9	-6,4	3,1%

Monthly revenue: actual (black), hold-out check (light blue), forecast Nov-Dec (blue), target (grey)



Full-year 2026 by scenario, EUR m (dashed = target EUR 120.9 m)



Full-year forecast by country (hold-out error per country)

Country	Forecast	Target	Gap	vs target	Error
Germany	23 234 374	25 681 254	-2 446 880	-9,5%	4,2%
France	24 880 742	27 179 984	-2 299 242	-8,5%	5,2%
United Kingdom	5 959 625	6 802 688	-843 063	-12,4%	10,7%
Italy	11 927 996	12 394 023	-466 028	-3,8%	2,6%
Netherlands	3 245 441	3 667 921	-422 480	-11,5%	13,6%
Australia	4 764 565	5 149 457	-384 892	-7,5%	3,1%
Belgium	4 004 683	4 258 421	-253 738	-6,0%	3,0%
Switzerland	3 943 262	4 139 286	-196 023	-4,7%	2,8%
Sweden	1 140 694	1 271 546	-130 852	-10,3%	6,5%

Hold-out check, May-Oct 2026

Model	Error, total	Error, median country	Bias
Naive: same month last year	1,8%	8,0%	-0,5%
Seasonal x trailing growth	3,1%	6,5%	3,1%

How to read it

The naive model (same month last year) has a lower total error (1.8%) but a higher error per country. Both say the year misses by about 5%.

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Claude got 29 of 30 test questions right and wrote 30 reviews with 0 untraceable numbers

SIMULATED DATA · AI

Deterministic work stays in SQL: every figure is computed first, then Claude only writes from those facts. A check extracts each number from the text and rejects the review if one is not in the facts. Questions in plain English become read-only SQL, and each answer shows its query. Reliability is measured, not assumed.

TEXT-TO-SQL · ACCURACY

29/30

core questions right; 87/90 over 3 runs, as the model is not deterministic.

TEXT-TO-SQL · STRESS TEST

10/10

harder questions: vague wording ('last quarter') or definitions not in the prompt.

REVIEWS · GROUNDING

0 / 1,670

numbers in 30 generated reviews not found in the SQL facts; 11/11 guard tests pass.

TIME SAVED · ESTIMATE

~46 h

saved a month, conservative: a person still reviews every output.

How it works

1 · Facts in SQL month vs target, gaps, price-volume-mix, at-risk hospitals: one JSON per review.
2 · Claude writes a one-page review from that JSON only (prompt in ai/prompts/).
3 · Number check any figure not in the JSON rejects the draft: one retry, then a plain template.
4 · Questions plain English to one SELECT; read-only database, file access off; the SQL is shown.

SQL or LLM? Numbers, joins, targets and forecasts: SQL and Python, always. The LLM writes and translates: summaries, plain-English questions. It never computes a figure that goes to management.

One command python ai/mbr_generator.py --month 2026-10 --all-countries writes 20 reviews.

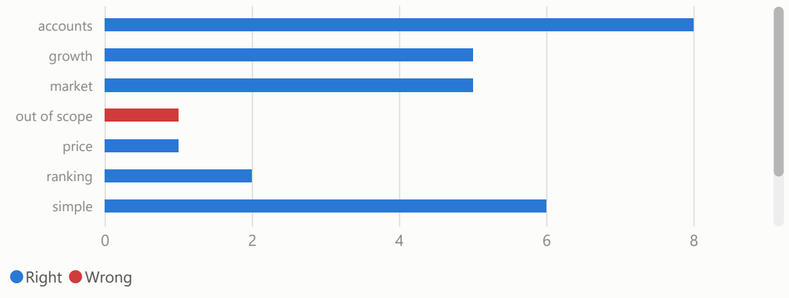
Monthly reviews generated for Jan-Oct 2026: numbers cited and numbers not found

Month	Numbers cited	Not found (draft)	Not found (published)	Accepted
2026-01	55	0	0	True
2026-02	62	0	0	True
2026-03	63	0	0	True
2026-04	62	0	0	True
2026-05	59	0	0	True
2026-06	68	0	0	True

Hours a month: by hand vs automated with review (assumptions in the README)

Task	By hand	Automated	Saved
Monthly business reviews (overall + 19 countries)	31,5	5,5	26,0
Ad-hoc sales questions	20,0	4,0	16,0
Account risk watch list	4,0	0,5	3,5
Total per month	55,5	10,0	45,5

Test questions by type (30 core + 10 stress), first run: right (blue), wrong (red)



Every wrong answer, read and explained

Q30 · What was our revenue in Japan last year?
Out of scope, failed in 3 of 3 runs. Japan is not in the data, but the prompt does not list the 19 countries, so Claude wrote a correct-looking query that returns NULL instead of saying 'cannot answer'. A NULL shown as 'no revenue' would mislead a reader. Fix: list the covered countries in the prompt and treat an empty result as 'not covered' in code. Not applied, so the score is not tuned on the test set.